Lecture 3, Jan 26 - 2026.

Rmk. We know that $f(x)=e^x$

$$\frac{d}{dx}(e^x)=e^x \quad \text{and} \quad \int e^x dx = e^x + C.$$

Properties of the Natural Exponential function: The exponential function $f(x)=e^x$ is an increasing continuous function, $\text{Dom} f=\mathbb{R}$ and $\text{range} f=(0,\infty)$. Thus $e^x>0$ for all $x\in\mathbb{R}$. Also

$$\lim_{x\to-\infty} e^x = 0, \qquad \lim_{x\to\infty} e^x = \infty.$$

So, the x-axis is a horizontal asymtote of $f(x)=e^x$.

Example. Find $\lim_{x\to\infty} \frac{e^{2x}}{e^{2x}+1}$.

Sln.

$$\lim_{x\to\infty} \frac{x^2+2x+1}{x^2+2x} = \lim_{x\to\infty} \frac{\frac{x^2}{x^2}+\frac{2x}{x^2}+\frac{1}{x^2}}{\frac{x^2}{x^2}+\frac{2x}{x^2}}$$

$$= \lim_{x\to\infty} \frac{1+2/x+1/x^2}{1+2/x} \qquad \left(\frac{\#}{\infty}=0\right)$$

$$= 1.$$

$$\lim_{x\to\infty} \frac{e^{2x}}{e^{2x}+1} = \lim_{x\to\infty} \frac{e^{2x}/e^{2x}}{e^{2x}/e^{2x}+1/e^{2x}}$$

$$= \lim_{x\to\infty} \frac{1}{1+e^{-2x}}.$$

$$= \frac{\lim_{x\to\infty} 1}{\lim_{x\to\infty}(1+e^{-2x})} = \frac{1}{1+\lim_{x\to\infty} e^{-2x}}.$$

$$= \frac{1}{1 + \lim_{x \to -\infty} e^{2x}} = \frac{1}{1+0} = 1.$$

Example. Use the first and second derivatives of $f(x) = e^{1/x}$, together with asymtotes to sketch its graph.

Sln Notice that $\text{Dom} f = \mathbb{R} - \{0\} = (-\infty, 0) \cup (0, \infty)$.

So, we check for vertical asymtote at $x = 0$.

$$\lim_{x \to 0^+} e^{1/x} = e^{\lim_{x \to 0^+} 1/x} = \infty \quad \text{b/c} \quad 1/x \to \infty \text{ as } x \to 0^+$$

then $e^{1/x} \to \infty$ as $x \to 0^+$.

this shows that $x = 0$ is a vertical asymtote.

$$\lim_{x \to 0^-} e^{1/x} = e^{\lim_{x \to 0^-} 1/x} = 0 \quad \text{b/c} \quad \frac{1}{x} \to -\infty \text{ as } x \to 0^-$$

then $e^{1/x} \to 0$ as $x \to 0^-$.

As $x \to \pm\infty$ we have $1/x \to 0$

$$\lim_{x \to \pm\infty} e^{1/x} = e^0 = 1.$$

this shows that $y = 1$ is an H.A.

Now, by the Chain-Rule:

$$f'(x) = \frac{d}{dx}(e^{1/x}) = e^{1/x} \cdot \frac{d}{dx}(1/x) = -\frac{e^{1/x}}{x^2}.$$

Since $e^{1/x} > 0$, $x^2 > 0$ for all $x \neq 0$ then $f'(x) < 0$ for all $x \neq 0$.

Thus f is decreasing on $(-\infty, 0) \cup (0, \infty)$. There is no critical numbers, so the function has no local max/min.

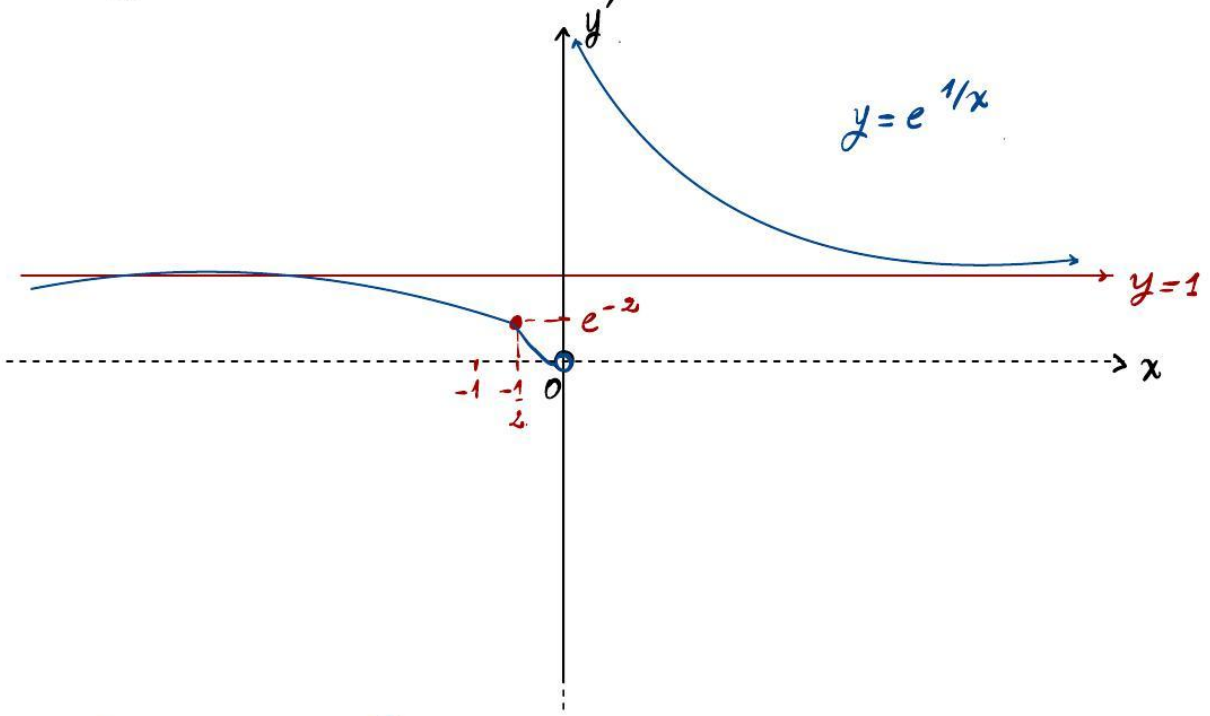
$$f''(x) = \frac{e^{1/x}(1/x^2) \cdot x^2 - (-e^{1/x}) 2x}{x^4} = \frac{e^{1/x}(2x+1)}{x^4}.$$

Since $e^{1/x} > 0$, $x^4 > 0$ $f''(x) > 0 \iff x > -1/2$ and $f''(x) < 0$

$\Leftrightarrow \quad x < -\frac{1}{2}$. So, f is $\cup$ on $(-\frac{1}{2}, 0) \cup (0, \infty)$ and

f is $\cap$ on $(-\infty, -\frac{1}{2})$.

at $x = -\frac{1}{2}$ we have inflution point its coordinates $(-\frac{1}{2}, e^{-2})$



6.3 Logarithmic Functions.

- **Logarithmic functions and their properties.**

If $b > 0$ and $b \neq 1$ the exponential function $f(x) = b^x$ is either increasing or decreasing and so it's one-to-one by the horizontal Line test. It therefore has an inverse function f^{-1}, which is called the logarithmic function with base b. and is denoted by $\log_b$

$$f^{-1}(x) = y \quad \overset{def}{\Leftrightarrow} \quad f(y) = x.$$

then, we have.

$$\log_b (x) = y \quad \Leftrightarrow \quad b^y = x$$

where, $x > 0$.

Example. $\log_3(81) = 4$ b/c $3^4 = 81$.

$\log_{25} 5 = \frac{1}{2}$ b/c $25^{1/2} = 5$.

Properties:

- $\log_b(b^x) = x$ for all $x \in \mathbb{R}$.
- $b^{\log_b(x)} = x$ for all $x \in \mathbb{R}$.

1. $\log_b(xy) = \log_b(x) + \log_b(y)$.
2. $\log_b\left(\frac{x}{y}\right) = \log_b(x) - \log_b(y)$.
3. $\log_b(x^r) = r \cdot \log_b(x)$.

Example. Use the laws of logorithms to evaluate:

a). $\log_4(2) + \log_4(32)$ b) $\log_2(80) - \log_2(5)$.

Sln. a) $\log_4(2) + \log_4(32) = \log_4(2 \times 32) = \log_4(64) = \log_4(4^3)$

$= 3 \underbrace{\log_4(4)}_{1.} = 3.$

b). $\log_2(80) - \log_2(5) = \log_2\left(\frac{80}{5}\right)$

$= \log_2(16) = \log_2(2^4) = 4 \underbrace{\log_2(2)}_{1} = 4.$

Rmk. if $b > 1$ then $f(x) = \log_b(x)$ is a one-to-one function, continuous, increasing, $\text{Dom } f = (0, \infty)$, $\text{Range } f = \mathbb{R}$.

Thm. If $b > 1$ then

$\lim_{x \to \infty} \log_b(x) = \infty$ and $\lim_{x \to 0^+} \log_b(x) = -\infty$.

Example. Find $\lim_{x\to 0} \log_{10}(\tan^2(x))$.

Sln. As $x\to 0$ let $t = \tan^2(x) \longrightarrow \tan^2(0) = 0$ So, since $b=10>1$ $\log_{10}(t)$ is continuous then.

$$\lim_{x\to 0} \log_{10}(\tan^2(x)) = \lim_{t\to 0^+} \log_{10}(t) = -\infty.$$



- *The Natural Logarithmic function.*

Of all the possible bases b for logarithms, we can fix $b=e$. The logarithm with base e is called the natural Logarithm.

Notation: $\log_e x = \ln x$.

Some Properties:

- $\ln x = y \iff e^y = x$.
- $\ln(e^x) = x$.
- $e^{\ln x} = x$.

} are inverses functions.

- $\ln(e) = 1$.
- $x^r = e^{r\ln x}$.

Example. Find x if $\ln x = 5$.

Sln.

$$\ln x = 5 \iff e^5 = x.$$

Example. Solve the equation $e^{5-3x} = 10$.

Sln. We take logarithms of both sides of the equation

$$\ln(e^{5-3x}) = \ln(10)$$

$$(5-3x)\underbrace{\ln(e)}_{1} = \ln(10)$$

$$5-3x = \ln(10)$$

$$x = \frac{1}{3}(5-\ln(10)).$$

Example. Use the laws of logarithms to expand $\ln\left(\frac{x^2\sqrt{x^2+2}}{3x+1}\right)$.

Sln.

$$\ln\left(\frac{x^2\sqrt{x^2+2}}{3x+1}\right) = \ln\left(x^2\cdot\sqrt{x^2+2}\right) - \ln(3x+1).$$

$$= \ln(x^2) + \ln\left(\sqrt{x^2+1}\right) - \ln(3x+1).$$

$$= 2\ln(x) + \frac{1}{2}\ln(x^2+1) - \ln(3x+1).$$

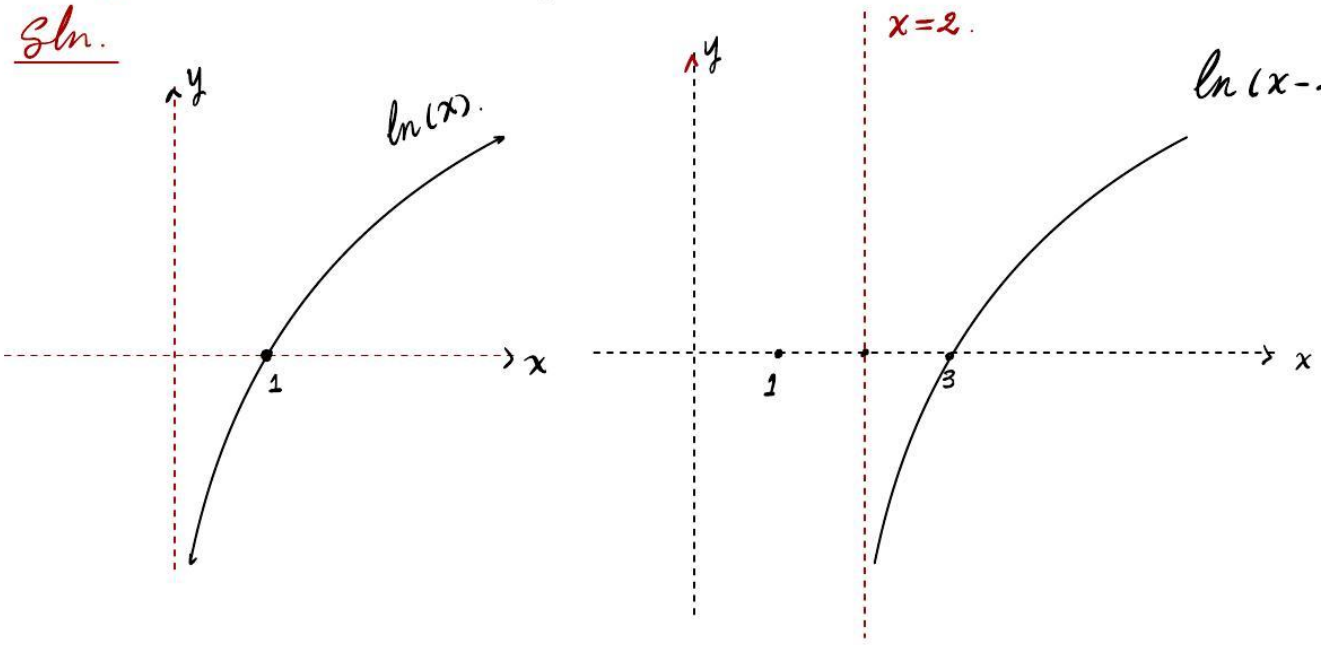
Prop. For any positive number $b \neq 1$ we have

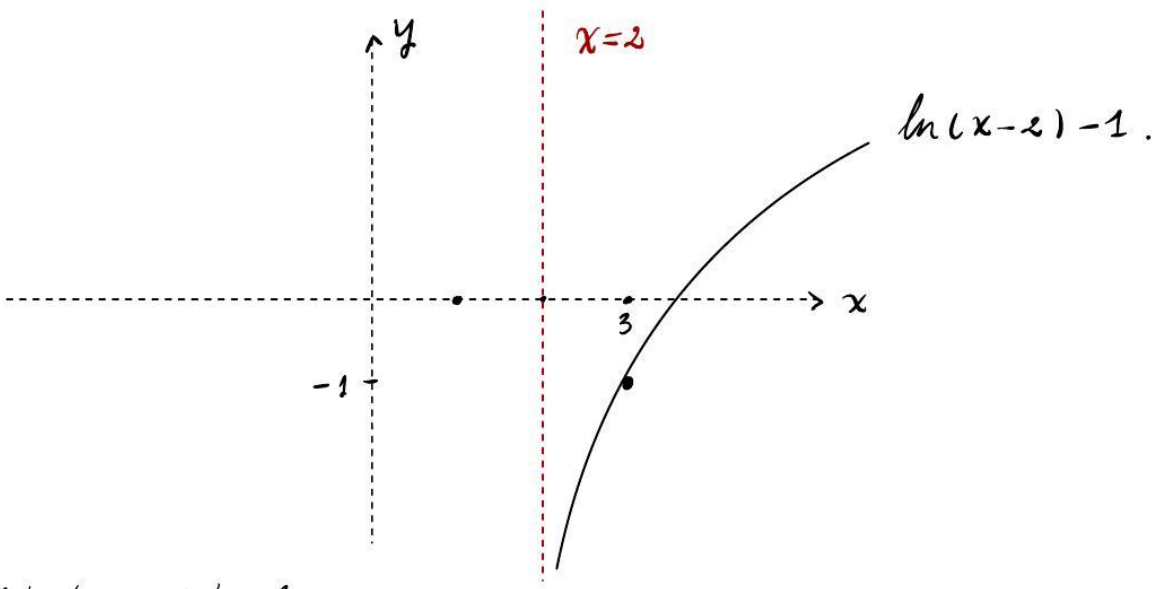
$$\log_b(x) = \frac{\ln(x)}{\ln(b)}.$$

Rmk. $\lim_{x\to\infty} \ln x = \infty$, $\lim_{x\to 0^+} \ln(x) = -\infty$, $\ln(e) = 1$.

Example. Sketch the graph of the function $y = \ln(x-2) - 1$

Sln.





Notice that the line $x=2$ is V.A.

$$\lim_{x \to 2^+} \left(\ln(x-2) - 1 \right) = -\infty.$$